Decision-time information in rolling airline recovery

An AOCC example: similar delay can imply different consequence profiles and recovery decisions.

Part 1. Decision-time operating state

Two disrupted aircraft-linked chains have similar delay at the AOCC decision epoch.

predecessor — turnaround — successor

factual milestones enter the state; future execution remains unresolved

Chain A

delay priority (P^D) — tight turnaround, aircraft continuity

Chain B

delay priority (P^D) — passenger connections, downstream exposure

State-informed consequence priority (P^C)

Flight, Passenger, and Resource consequences can distinguish chains that appear similar under delay alone.

Part 2. Stage I screening

Delay priority (P^D) versus consequence priority (P^C)

Which chains enter detailed review under limited capacity?

Chain A candidate

Chain B candidate

top-K review boundary

Part 3. Stage II local recovery

For a shortlisted chain: (u=0) or (u>0)?

The target is selected under a consequence-effort objective.

Feasible target set

(u=0): no recovery — (u>0): positive target

Information value is observed through shortlist or target changes.